

Title - FederiGene: Federated Genomic Intelligence for Secure Multi-Institutional Disease Classification

P - Problem

Genomic data required for accurate disease prediction is distributed across multiple hospitals and research institutions. Due to privacy regulations, ethical constraints, and data security risks, raw patient genomic data cannot be centrally pooled for machine learning model training. This results in fragmented datasets, limited model generalization, and reduced predictive performance in precision medicine applications.

I - Intervention

Implementation of FederiGene, a privacy-preserving federated learning framework that enables multiple institutions to collaboratively train genomic disease prediction models without sharing raw patient data. The system uses distributed local model training, secure weight aggregation, and integrity verification mechanisms to ensure data privacy and model reliability.

C - Comparison

Traditional centralized machine learning approaches where genomic datasets from multiple institutions are pooled into a single server for training, increasing risks related to data privacy, regulatory compliance, and potential data breaches.

O - Outcome

Primary Outcomes:

- Comparable or improved disease classification accuracy relative to centralized models.
- Preservation of patient genomic data privacy (no raw data transfer).

Secondary Outcomes:

- Secure and integrity-verified model updates.
- Reduced risk of data exposure.
- Feasible multi-institutional collaboration for precision medicine.